



Pre-Junior Cycle Final Examination, 2024

Mathematics

Higher Level

Time: 2 hours

270 marks

CANDIDATE DETAILS

**DAY and MONTH
of BIRTH**

____ / ____

For example, 3rd February
is entered as 03/02

NAME [illegible]

SCHOOL

[illegible]

TEACHER 

School Stamp

For Examiner only

Question	Mark
1.	
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Total	
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Instructions

There are 11 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. You may ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1**(Suggested maximum time: 5 minutes)**

- (a) Write each of the following numbers in the form $a\sqrt{2}$, where $a \in \mathbb{N}$.

$$\sqrt{8}, \quad \sqrt{18}, \quad \sqrt{32}$$

Use your answers to find the value of $\sqrt{8} + \sqrt{32} - \sqrt{18}$, giving your answer in surd form.

- (b) The average human hair has a diameter of 0.008 cm.
Write 0.008 cm in the form $a \times 10^n$, where $1 \leq a < 10$ and $n \in \mathbb{Z}$.

- (c) Using the laws of indices, solve the equation

$$\frac{5^m \times 5^7}{5^{21}} = 1$$

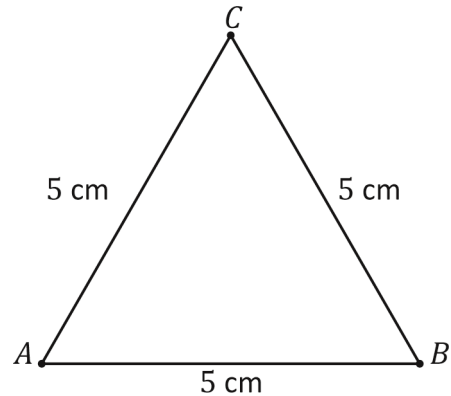
to find the value of m .



Question 2 (Suggested maximum time: 5 minutes)

Question 2 (Suggested maximum time: 5 minutes)

- (a)** A triangle ABC has sides of length 5 cm as shown.



- (i) What type of triangle is shown?

Answer =

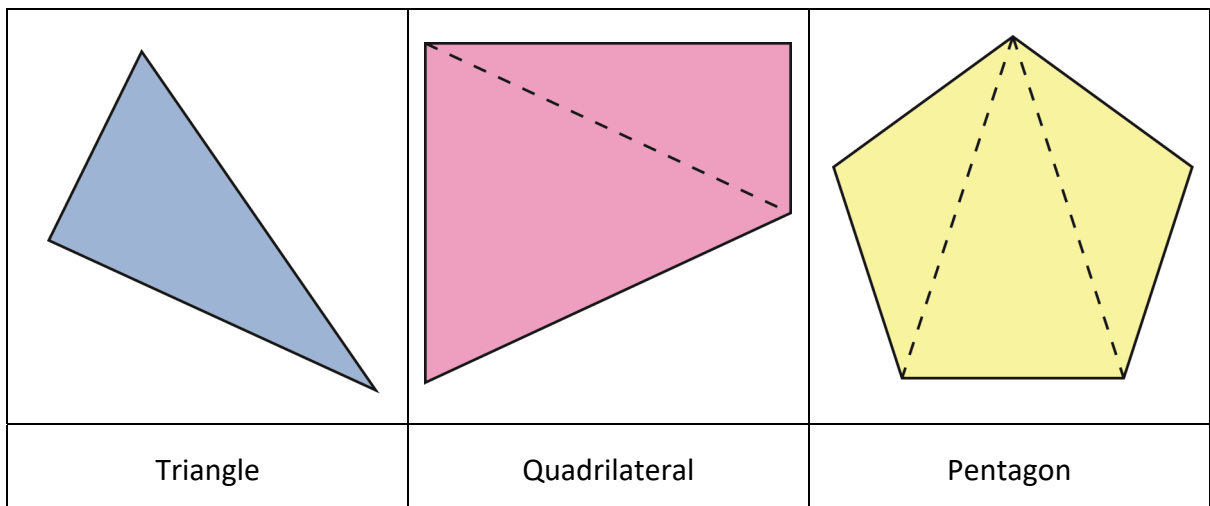
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- (ii) Write down the size of the angle $\angle ABC$.

[illegible]

- (b)** Each of the following shapes can be divided into a number of triangles, as shown.

A group of students discovered that there is also a relationship between the number of sides in each shape and the sum of the shape's interior angles.



- (i) Fill in the missing values in the table below, showing the number of sides, the number of triangles formed and the sum of the interior angles for each shape.

Shape	Number of sides, n	Number of triangles formed	Sum of the interior angles
Triangle	3	1	180°
Quadrilateral		2	
Pentagon	5		

[illegible]

- (ii) Tick (✓) the correct box to show which algebraic expression below represents the sum of the interior angles in a shape with n sides, where $n \geq 3$ and $n \in \mathbb{N}$. Tick **one** box only. Justify your answer.

$$180n$$

1

$$180(n - 2)$$

$$180(n + 2)$$

1

Justification:

Justification:	
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- (iii)** Work out the sum of the interior angles in an octagon, an 8-sided shape.

[illegible]

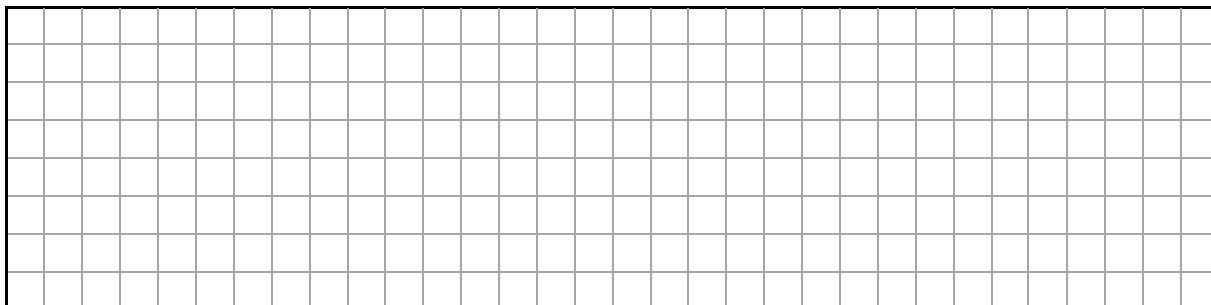
Question 3

(Suggested maximum time: 10 minutes)

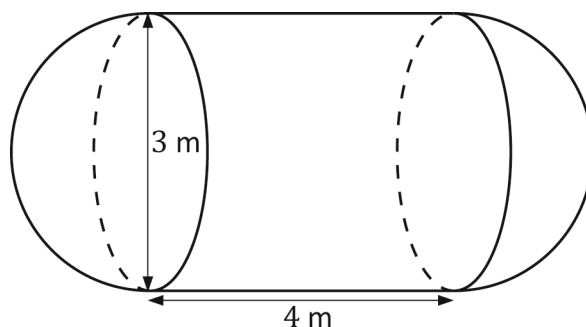
A circular Ferris wheel has a circumference length of 125π metres.



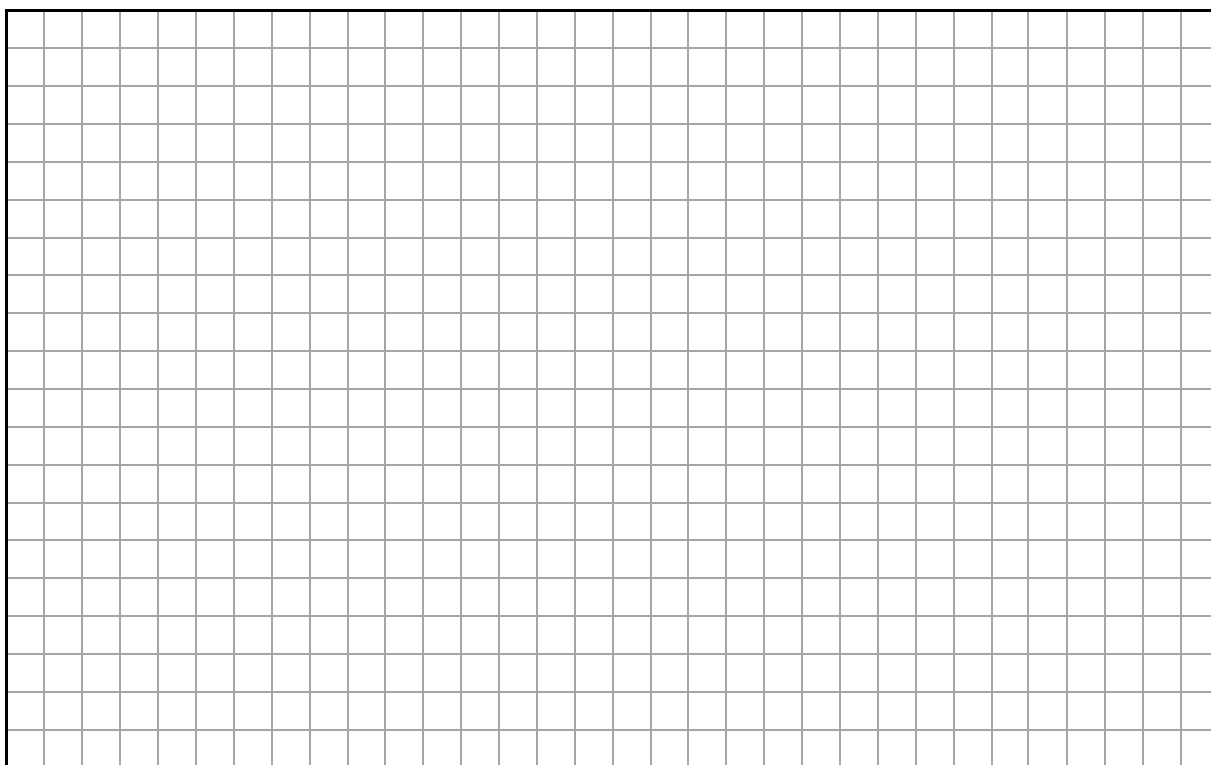
- (a) Work out the length of the **radius** of the Ferris wheel.



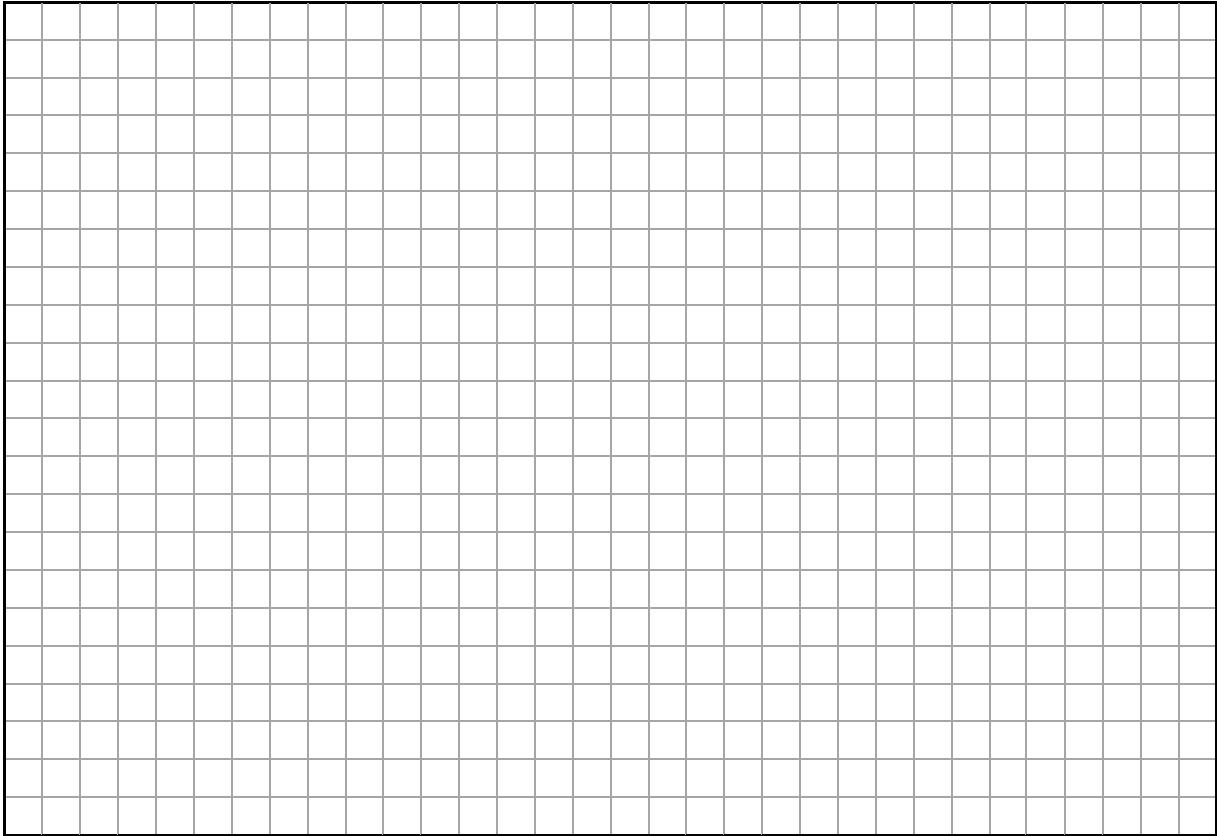
- (b) There are 30 pods on the Ferris wheel. Each identical pod is made of glass. They are in the shape of a cylinder with a hemisphere at each end. The cylindrical part has a diameter of 3 m and a height of 4 m.



- (i) Work out the **volume** of each pod in terms of π .
Write your answer in the form $\frac{a}{b}\pi \text{ m}^3$, where $a, b \in \mathbb{N}$.

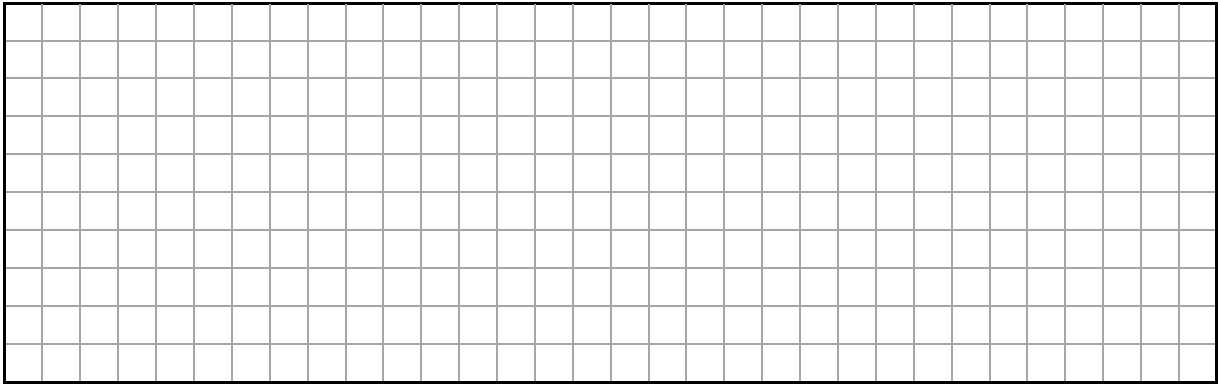


- (ii) Work out the **total surface area** of glass required to make all 30 pods.
Give your answer correct to one decimal place.

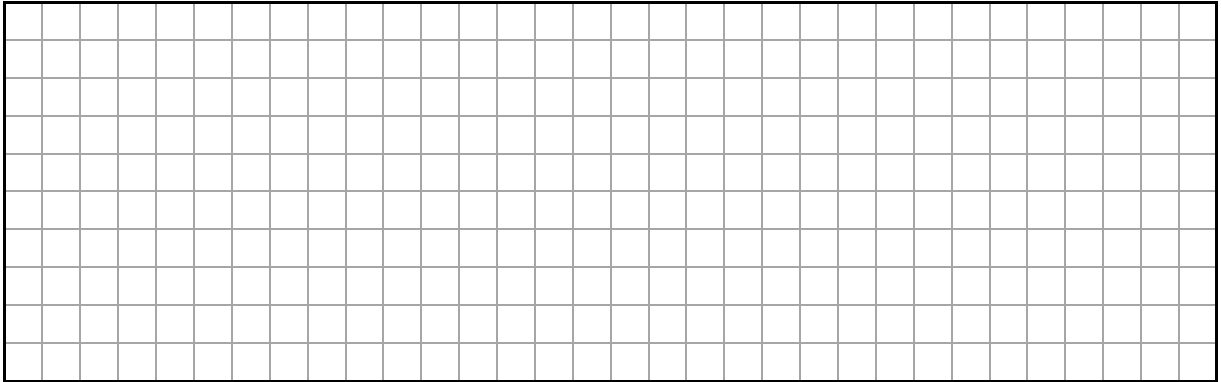


Question 4**(Suggested maximum time: 5 minutes)**

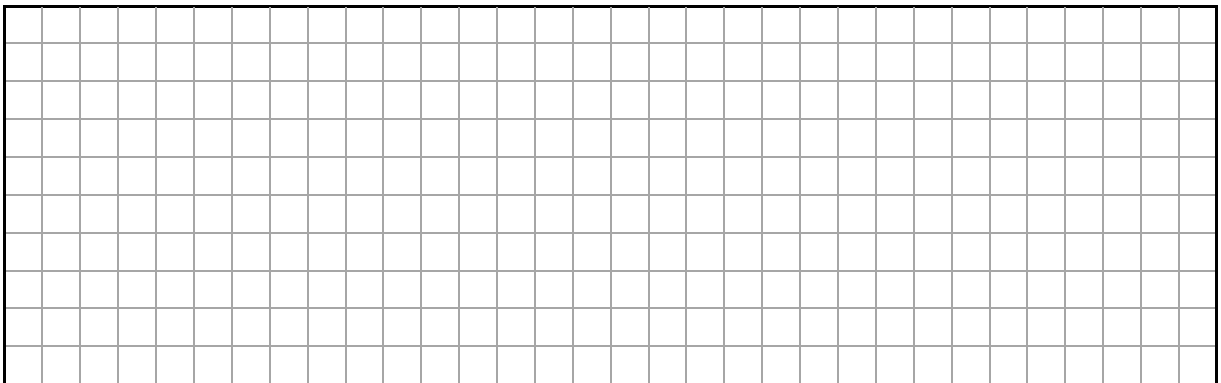
- (a) $a = -5$ and $b = \frac{1}{9}$. Work out the value of $a^2 - 6ab$.



- (b) Factorise fully $8x^2 - 6x - 4xy + 3y$.



- (c) Solve the equation $3(5x - 9) + 8 = 12x - 1$.



Question 5 (Suggested maximum time: 10 minutes)

Question 5 (Suggested maximum time: 10 minutes)

Luca has a gross annual income of €54 000.

Luca pays income tax on his gross income at a rate of 20% on the first €42 000, and 40% on the balance.

- (a)** Work out Luca's annual income tax at each of these two rates (20% and 40%).

Income tax at 20% = € _____ Income tax at 40% = € _____

- (b)** Luca has annual tax credits of €4150.
Work out Luca's annual take-home pay, after income tax has been deducted.



- (c) Luca receives a pay rise. His annual take-home pay, after income tax has been deducted, is now €49 000. Work out Luca's new gross annual income, correct to the nearest euro.

[illegible]

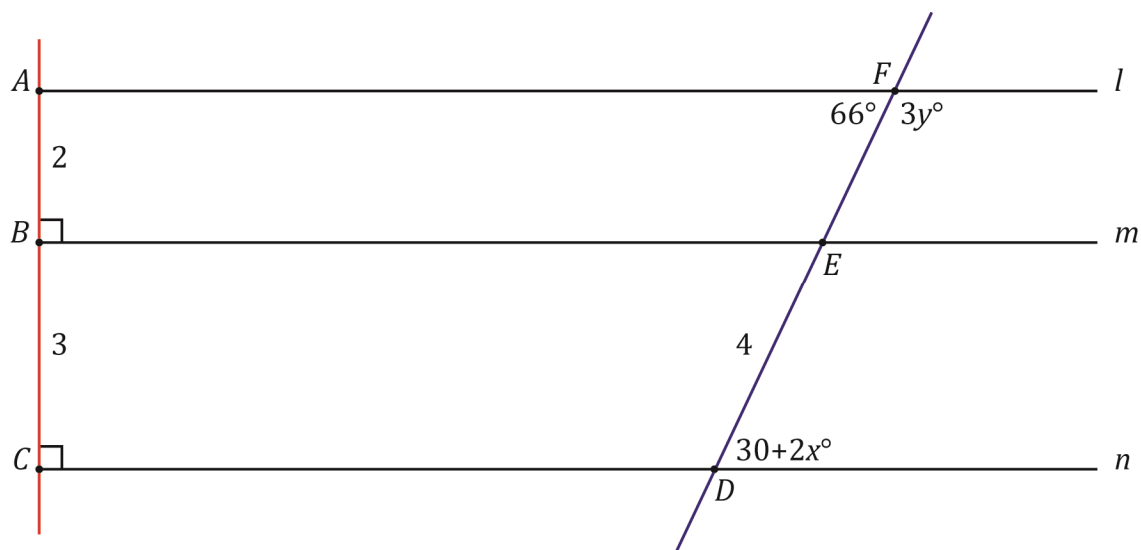
Question 6

(Suggested maximum time: 10 minutes)

The diagram shows three parallel lines l , m and n and two transversals (not to scale).

The transversal $[AC]$ is perpendicular to l , m and n .

The lengths of some of the sides and measures of some of the angles are given in the diagram, with $x, y \in \mathbb{N}$.



- (a) Write down two alternate angles from the diagram.

Alternate angles:

and

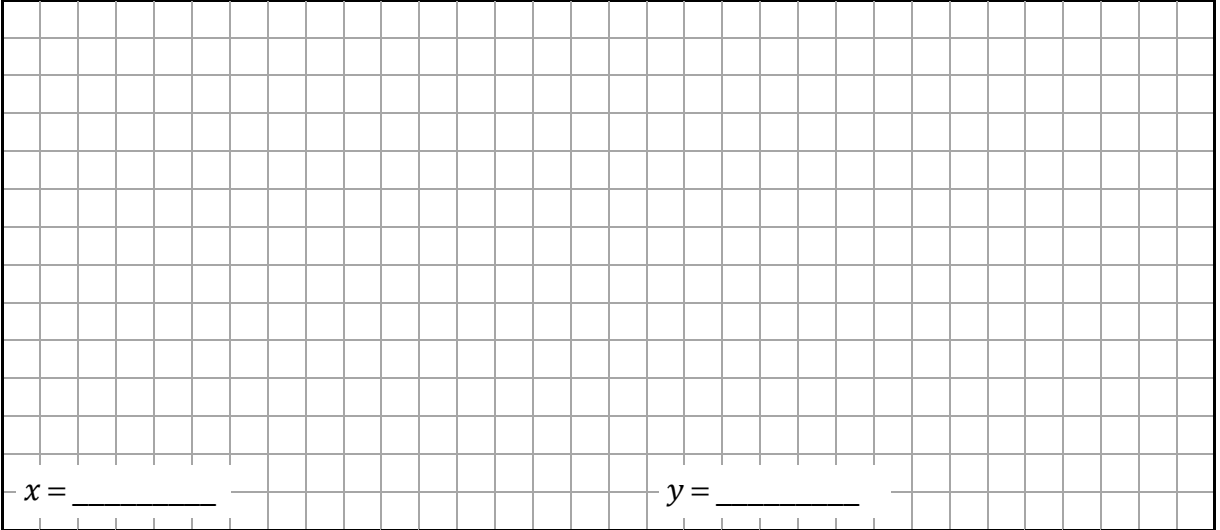
- (b) Write down two angles from the diagram that sum to 180° .

+

$= 180^\circ$

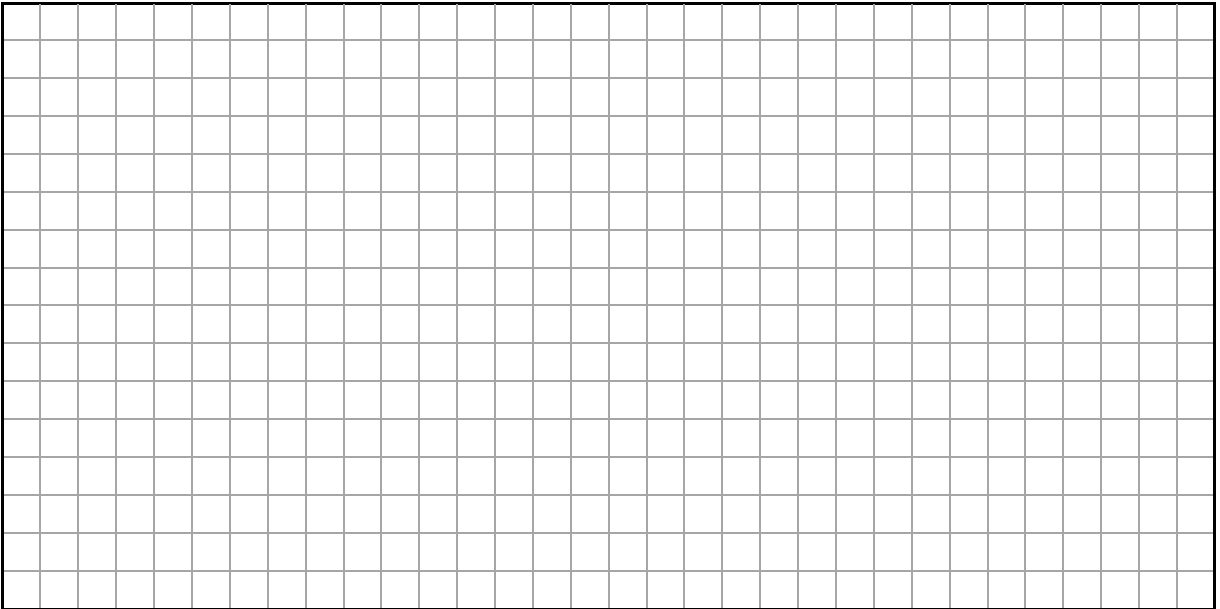


- (c) Work out the value of x and the value of y in the diagram.



$x =$ _____ $y =$ _____

- (d) Work out the length of the line segment $[EF]$.



(Suggested maximum time: 15 minutes)

-

Key: | 1 | 2 = 12 people

- 7

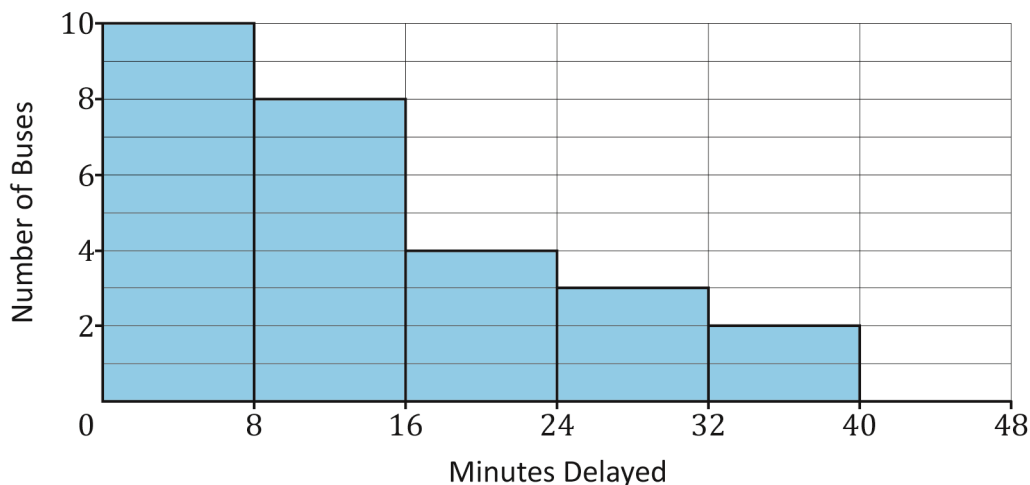
[illegible]

- [illegible]

- | |
|--|
| |
|--|

- [illegible]

- (b)** The delay (in minutes) by buses reaching their final destination in the city was also recorded. The buses delayed by between 0 and 40 minutes are shown in the histogram below.



- (i) Use the information in the histogram to complete the table below.

Minutes delayed	0 – 8	8 – 16	16 – 24	24 – 32	32 – 40	40 – 48
Number of buses	10		4			y

[Note: 8 – 16 means “a delay of 8 minutes or more but less than 16 minutes”, etc.]

- (ii) The estimated **mean** delay of the buses is 18 minutes and 30 seconds.
Use **mid-interval values** to find the value of γ in the table above, where $\gamma \in \mathbb{N}$.

[illegible]

- (iii)** Use your value of γ to complete the histogram above.



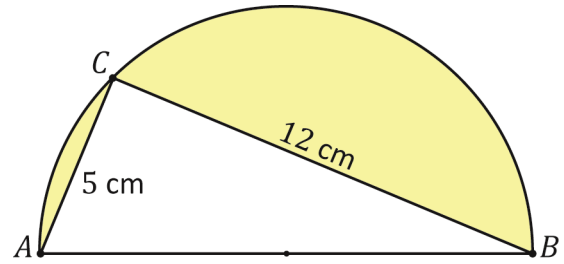
Question 8

(Suggested maximum time: 15 minutes)

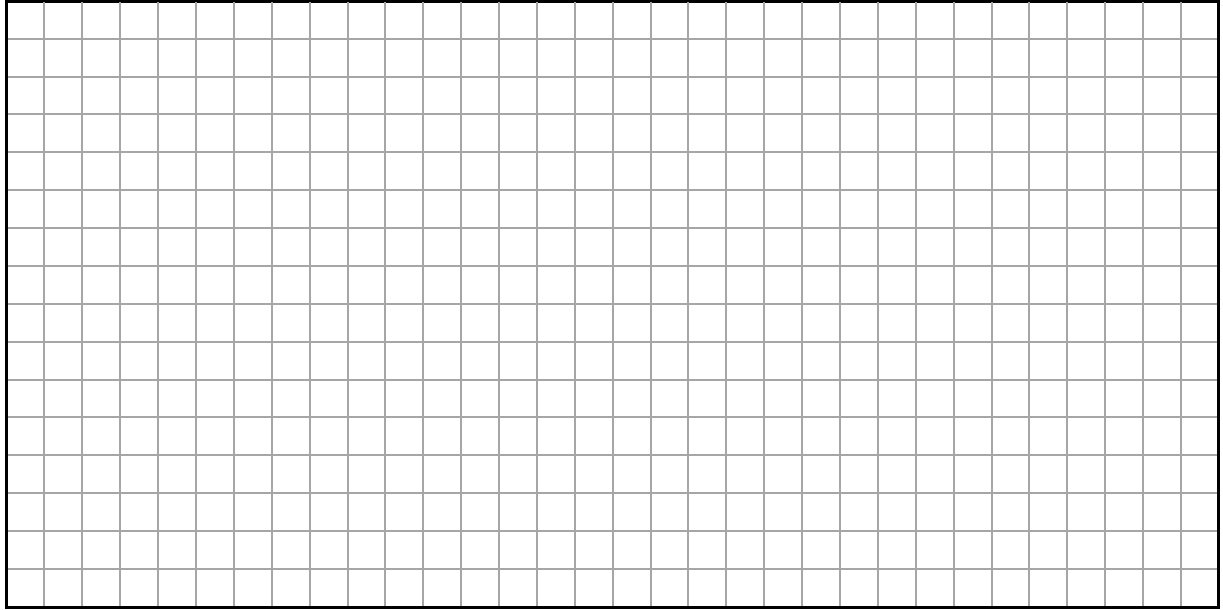
The vertices of the triangle ABC touch a semicircle as shown in the diagram.

$[AB]$ is the diameter of the semicircle.

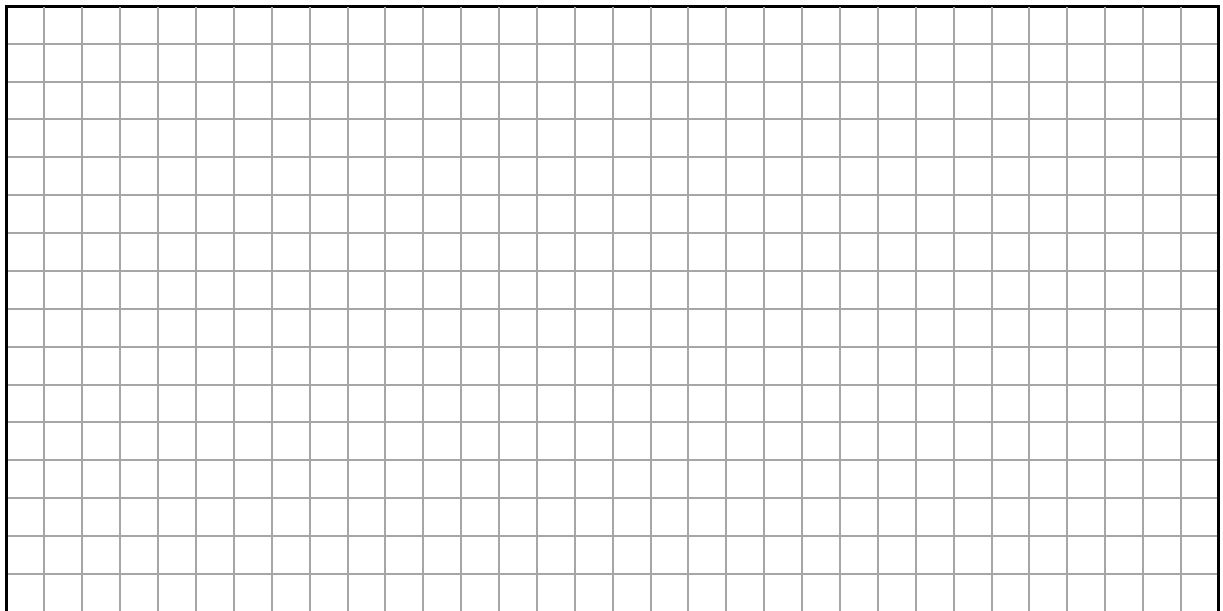
$|AC| = 5 \text{ cm}$ and $|BC| = 12 \text{ cm}$.



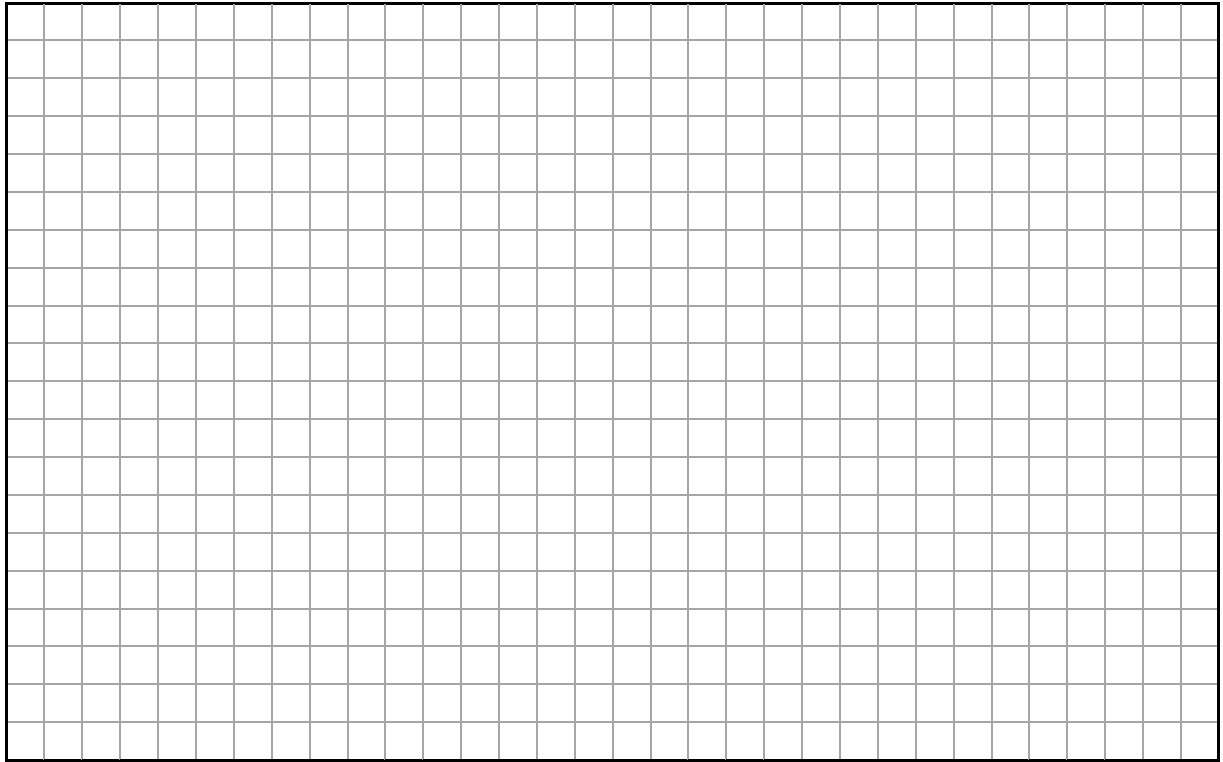
- (a) Show that the semicircle has a radius of 6.5 cm .



- (b) Use trigonometry to work out the size of the **smallest angle** in the triangle ABC .
Give your answer in degrees, correct to one decimal place.



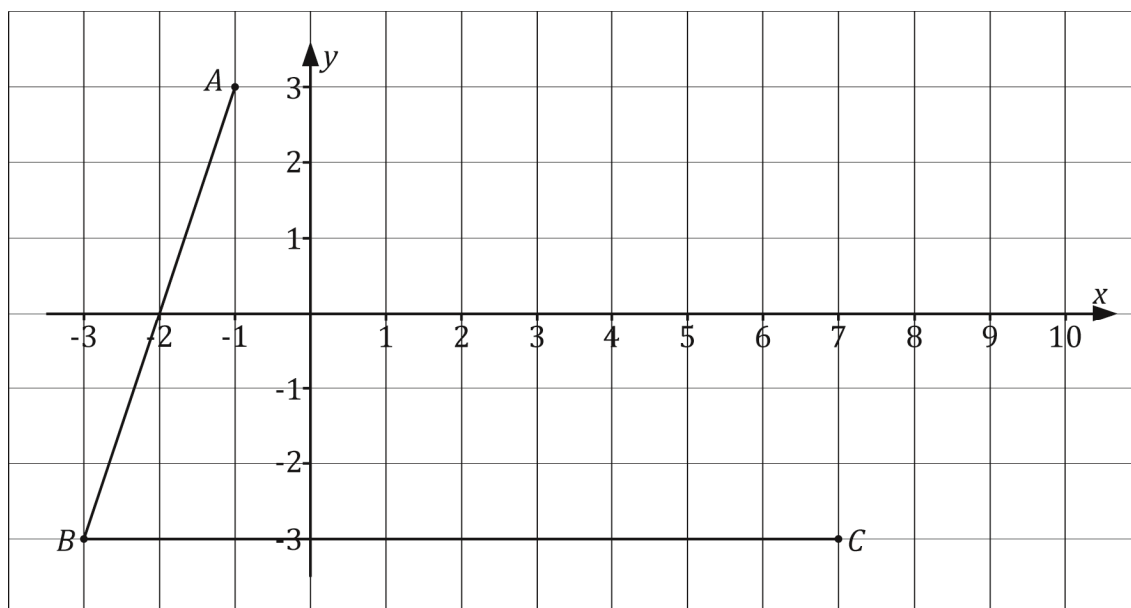
- (c) Find the **area** of the shaded region in the diagram.
Express this area as a **percentage** of the total area of the semicircle.
Give your answer correct to one decimal place.



Question 9

(Suggested maximum time: 15 minutes)

A park, in the shape of a parallelogram $ABCD$, is being constructed as part of a new town plan. The map of the park is partially constructed, with the points A , B and C shown on the co-ordinate diagram below.



- (a) Write the co-ordinates of the points A , B and C in the spaces below.

$A = (\quad , \quad)$

$B = (\quad , \quad)$

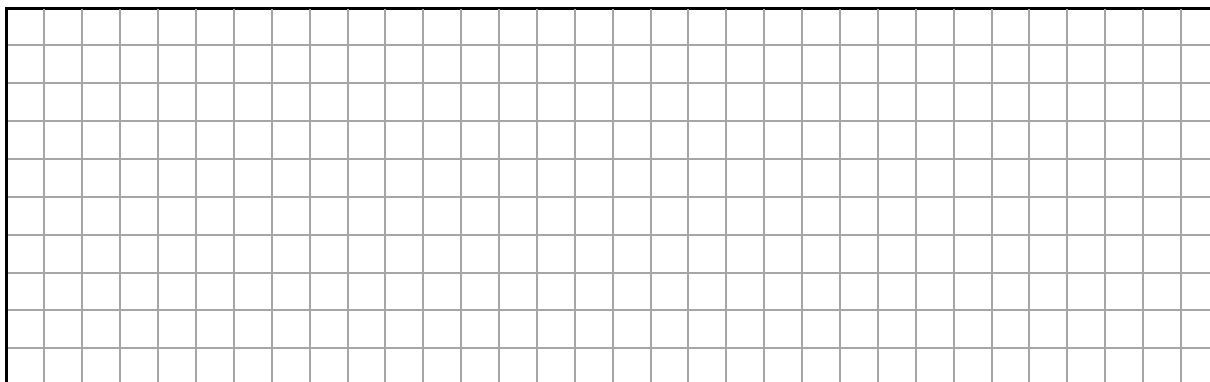
$C = (\quad , \quad)$

- (b) Complete the parallelogram to show the point D . Write down the co-ordinates of D .

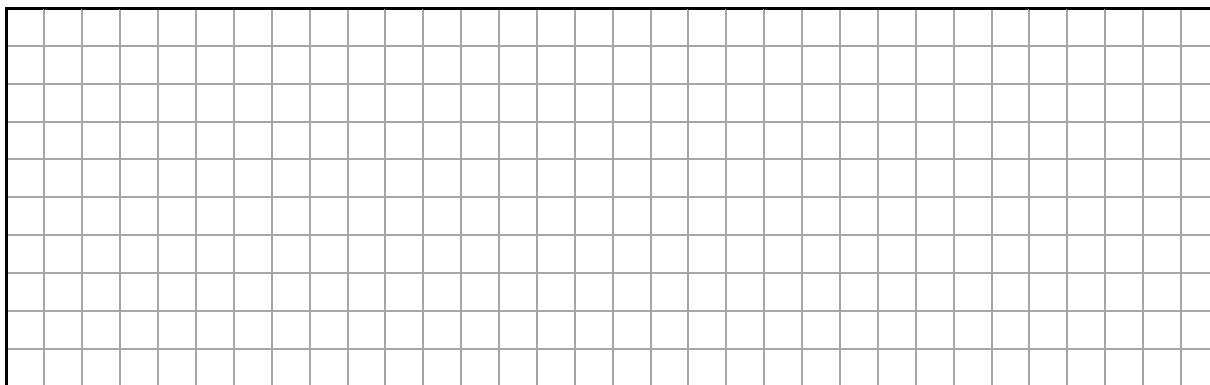
$D = (\quad , \quad)$

- (c) A fountain is to be built at the midpoint between the points A and C . Write down the co-ordinates of the fountain.

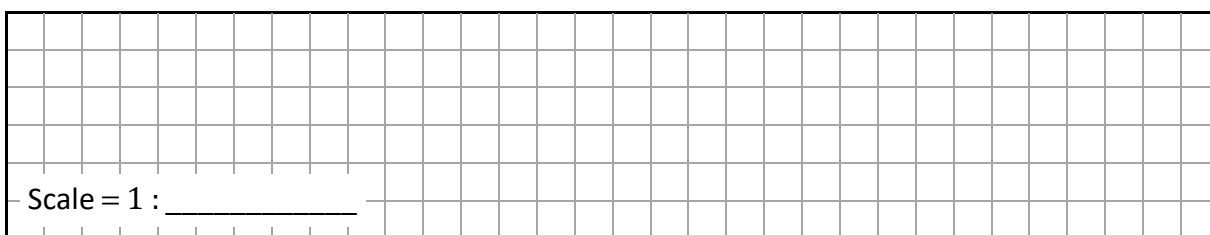
- (d) A fence is to be built around the perimeter of the park.
Find $|AB|$ and hence work out the length of the fence in the diagram.
Give your answer in cm, correct to two decimal places.



- (e) Each 1 cm unit on the co-ordinate diagram represents an actual distance of 50 m.
The actual area of the park is 150 000 m².
Work out the area of the parallelogram in the diagram.



- (f) Work out the scale of the diagram. Give your answer in the form $1 : n$, where $n \in \mathbb{N}$.

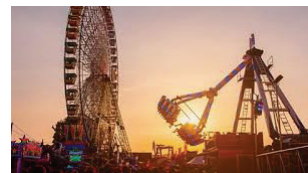


Question 10

(Suggested maximum time: 10 minutes)

100 students went on a trip to an amusement park.

Among the students, the three most popular rides at the park were the Mega rollercoaster (M), the Gravitator (G) and the Haunted Hotel (H).



- (a)** Complete the Venn diagram using the following data.

24% of the students went on all 3 of these rides.

$\frac{1}{25}$ of the students didn't go on any of these rides.

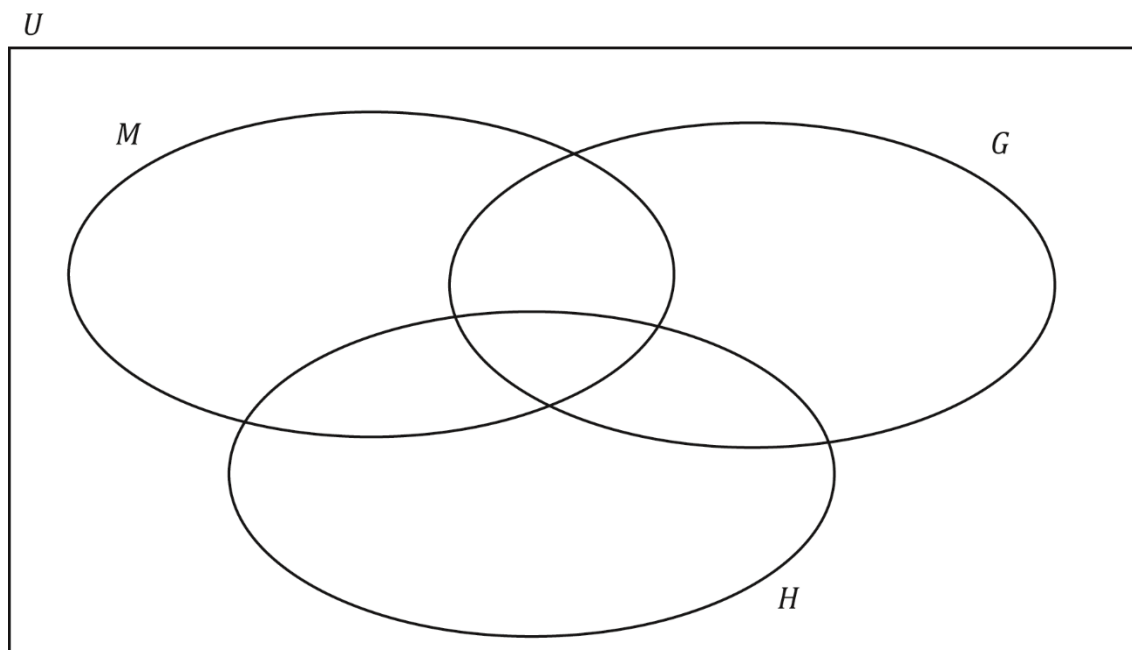
6 went on the Haunted Hotel only.

16 students went on the Mega rollercoaster and the Haunted Hotel but not the Gravitor.

17 students went on the Gravitator and the Haunted Hotel but not the Mega rollercoaster.

10 students went on the Mega rollercoaster and the Gravitor but not the Haunted Hotel.

$$\#(M \setminus (G \cup H)) = x \text{ and } \#(G \setminus (M \cup H)) = y.$$

[illegible]

- (b)** Find the **probability** that a student, picked at random, went on at least 2 rides.

[illegible]

- (c) The total number of students who went on the Mega rollercoaster is equal to the total number of students who went on the Gravitator.

Using the Venn diagram, write down two equations in x and y .

Equation 1: _____	Equation 2: _____
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- (d) Hence work out the value of x and the value of y , where $x, y \in \mathbb{N}$.

$x =$ _____	$y =$ _____
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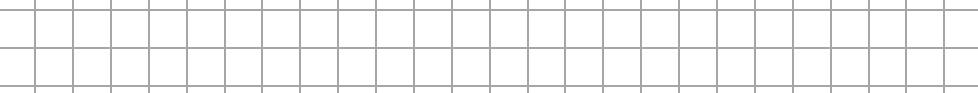
Question 11 (Suggested maximum time: 10 minutes)

Question 11 (Suggested maximum time: 10 minutes)

The first four terms of a sequence are as follows:

2, 8, 16, 26

- (a)** Work out **term 5**, the next term of the sequence.



- (b)** What type of sequence do these terms follow?
Tick (✓) the correct box below to show your answer.
Tick **one** box only. Justify your answer.

linear

☐

quadratic

exponential

5

Justification:

Justification:



- (c)** Work out an expression for the **n th term** of this sequence.

[illegible]

- (d)** Hence, or otherwise, work out the **20th term** of this sequence.

[illegible]

Page for extra work.

Label any extra work clearly with the question number and part.

[illegible]

Label any extra work clearly with the question number and part.

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Pre-Junior Cycle Final Examination, 2024

Mathematics – Higher Level

Time: 2 hours

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